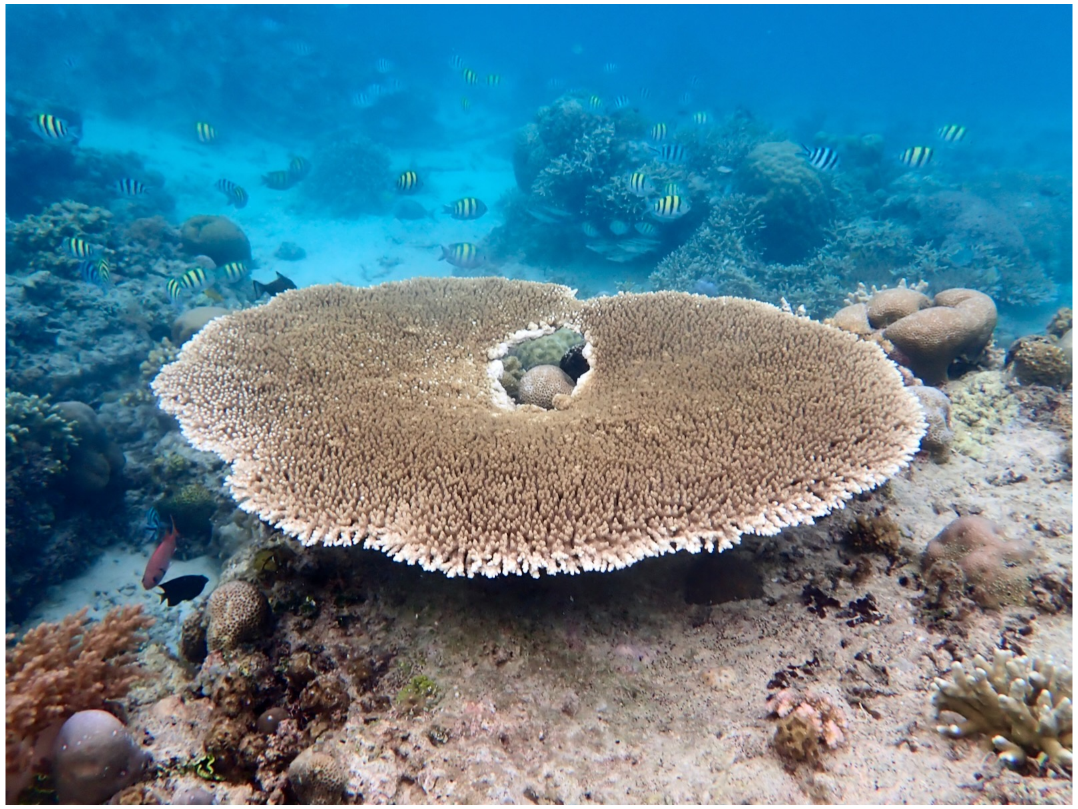
Exercises in Marine Ecological Genetics

06. Population genomics and genetic diversity

- Heterozygosity and F_{IS} using SNPs
- Nucleotide diversity
- Reduce linkage disequilibrium bias



Martin Helmkampf

https://github.com/mhelmkampf/meg26

Today's objectives

By the end of this session, you will be able to:

- Calculate and interpret heterozygosity and **inbreeding coefficients (F_{IS})** per individual from SNP data
- Estimate and compare **nucleotide diversity (π)** between populations using window-based approaches
- Describe the effect of linkage disequilibrium on population genomic datasets and apply **SNP thinning** to reduce it
- Use **shell loops** to apply population genomic analyses consistently and reproducibly across multiple samples, populations, or genomic subsets

Variant call format (VCF)

```
zcat < snps_hamlets_filtered.vcf.gz | head
```

```
##fileformat=VCFv4.1
##fileDate=02012019_20h38m04s
##source=SHAPEIT2.v837
##log_file=shapeit_02012019_20h38m04s_959049fa-700a-4d37-a4ff-3b5db0353190.log
##FORMAT=<ID=GT,Number=1,Type=String,Description="Phased Genotype">
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT 18158nigbel 18159nigbel 18162nigbel …
LG12 4152 . T G . PASS . GT 0|0 0|0 0|0 …
LG12 4228 . C A . PASS . GT 0|1 0|0 0|1 …
LG12 4262 . A G . PASS . GT 1|0 0|1 1|0 …
LG12 4263 . C T . PASS . GT 0|1 1|0 0|1 …
```

0|0 Homozygous for reference (1st) allele
1|1 Homozygous for alternate (2nd) allele

0|1 and 1|0 Heterozygous

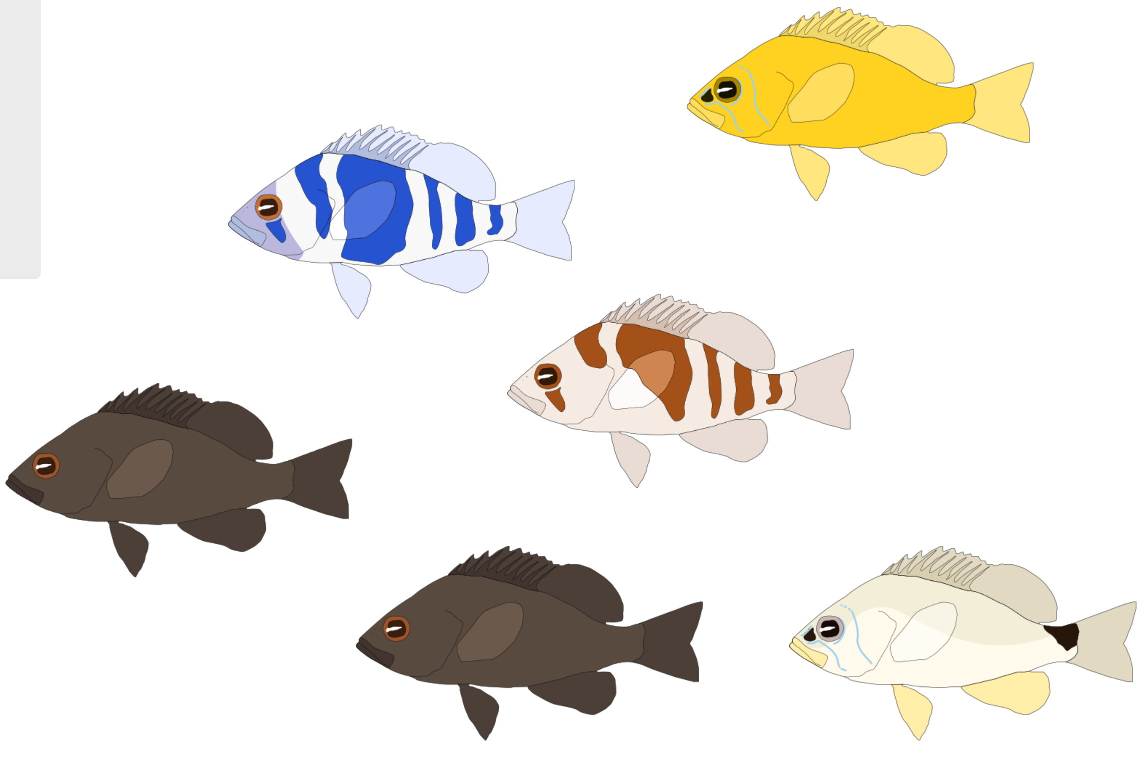
Hamlet SNP data

- 19 species of hamlet (genus *Hypoplectrus*)
- 15 sites in Caribbean and Gulf of Mexico
- 327 hamlet samples total
- Illumina short-read resequencing (mean depth 17×)
- Mapped against *H. puella* reference genome
- Genotyped with GATK

↓

hamlets_snps_lg12.vcf.gz

- Chromosome 12 only
- Subset to 36 samples from 6 populations

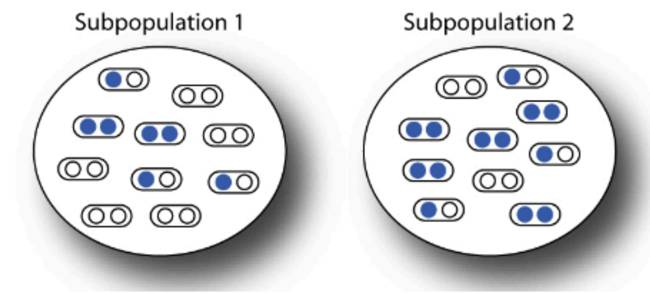


Illustrations by Kosmas Hench

F-statistics: multiple subpopulations

Exercise 1

In *n* subpopulations



$$H_I = \frac{1}{n}\sum_{i=1}^{n}\hat{H}_i$$ Mean observed heterozygosity within subpopulations

$$H_S = \frac{1}{n}\sum_{i=1}^{n}2p_iq_i$$ Mean expected heterozygosity within subpopulations (assuming HW within subpops)

$$H_T = 2\overline{p}\overline{q}$$ Expected heterozygosity of the total population (assuming HW within the total pop)

$\hat{H}_i$ observed heterozygosity in population i $p_i(q_i)$ frequency of allele A(a) in population i

$\overline{p}(\overline{q})$ mean frequency of allele A(a) n: number of subpopulations

$$F_{IS} = \frac{H_S - H_I}{H_S} \qquad F_{ST} = \frac{H_T - H_S}{H_T} \qquad F_{IT} = \frac{H_T - H_I}{H_T}$$

F_{IS} Average difference between observed and Hardy–Weinberg expected heterozygosity within each subpopulation (due to non-random mating)

F_{ST} Reduction in heterozygosity due to subpopulation divergence of allele frequencies

F_{IT} *Combined departure from HW expected genotype frequencies due to non-random mating within subpopulations and divergence of allele frequencies among subpopulations*

Nucleotide diversity π

Exercise 2

Average number of nucleotide differences per site between all possible pairs of sequences

$$\hat{\pi} = \frac{n}{n-1} \sum_{ij} x_i x_j \pi_{ij}$$

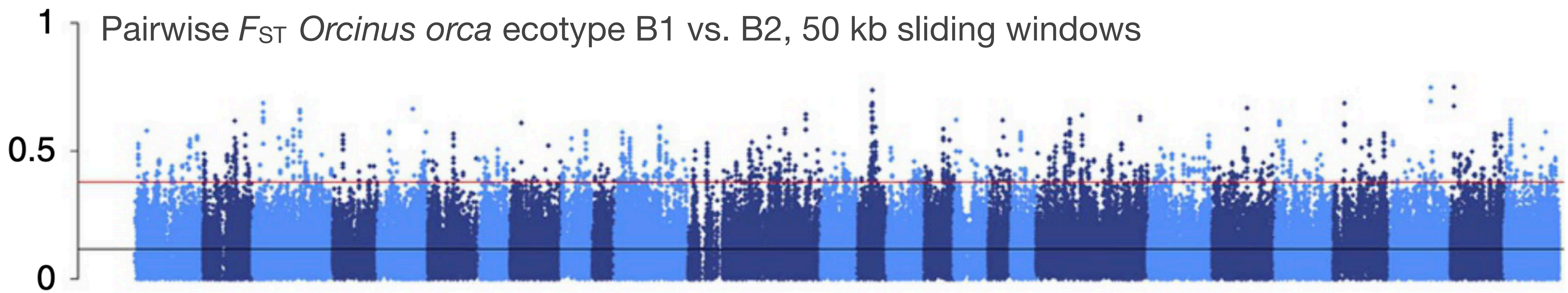
- 2 chromosomes per individual
- 15 pairwise comparisons
- π per site: no. diff / comparisons

Average π = 0.29

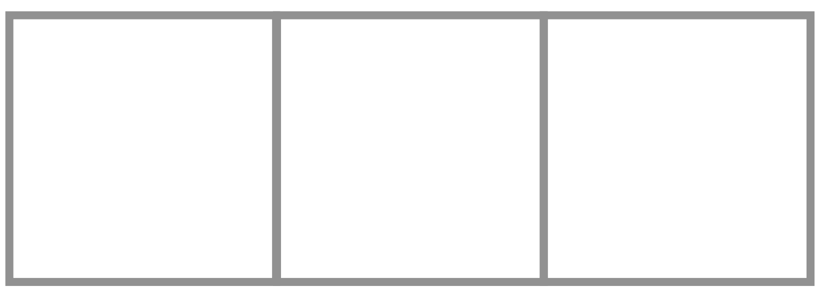
Site:	1	2	3	4	5
Sample A:	A	G	C	T	T
	A	G	C	T	T
Sample B:	A	G	T	T	T
	A	G	T	T	T
Sample C:	G	G	T	C	T
	G	G	C	T	T
No. diff:	8	0	9	5	0
π per site:	0.53	0	0.60	0.33	0

Genome-wide scans using windows

Exercise 2



Foote et al. 2016, *Nature Communications*



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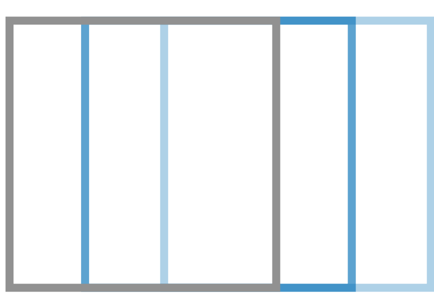
Non-overlapping windows



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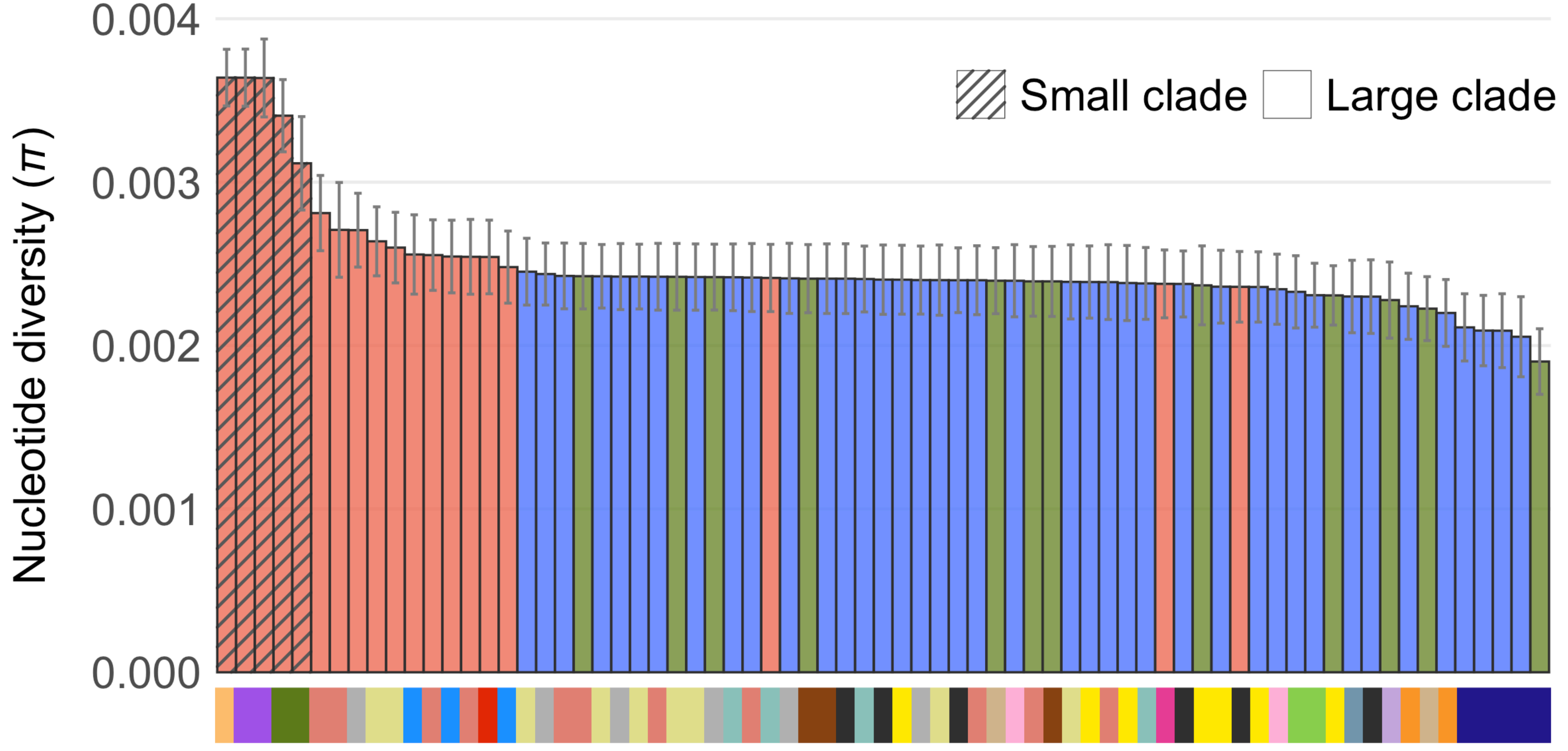
Overlapping / sliding windows

Window size

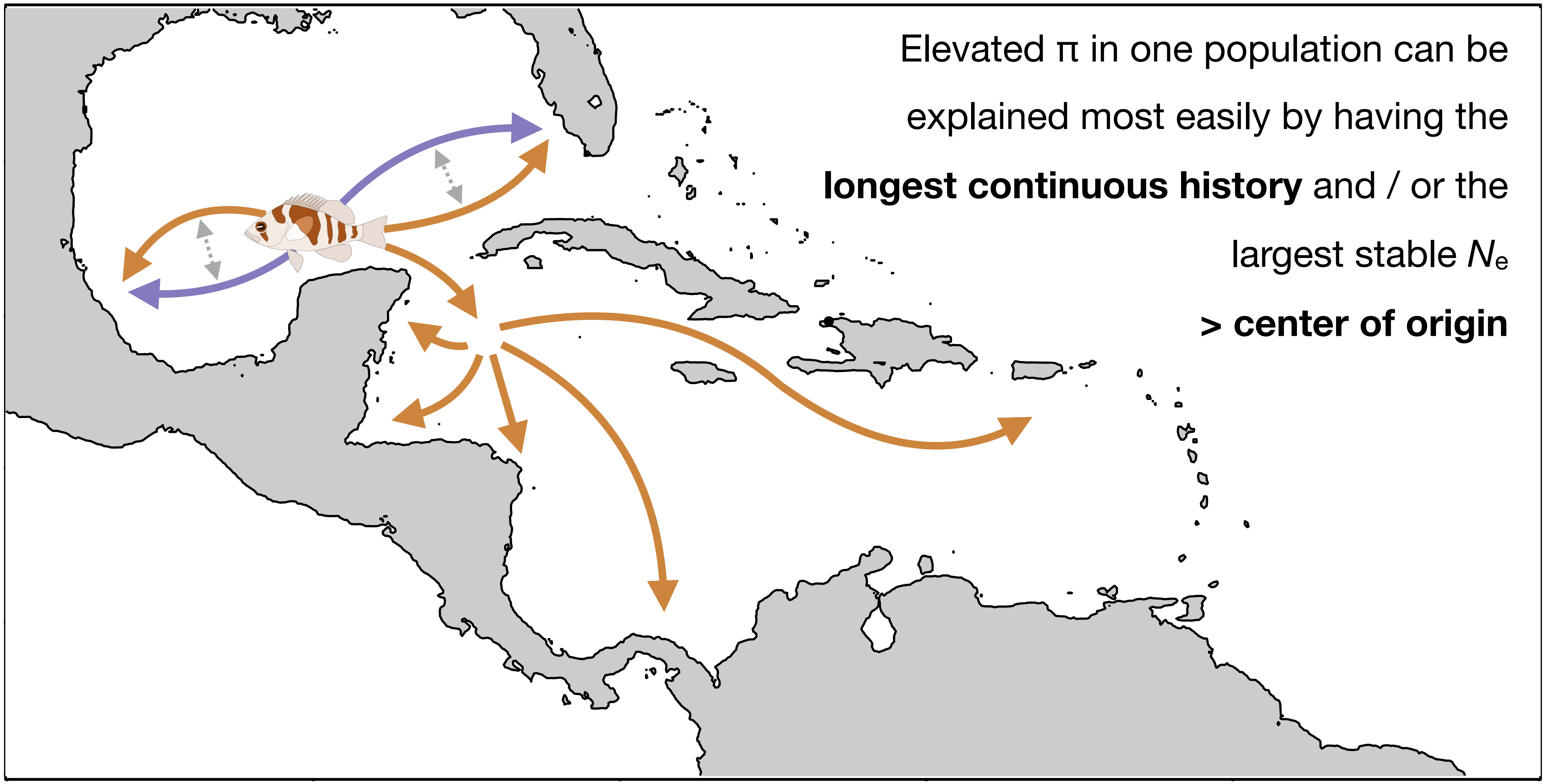
Increment / step size

Nucleotide diversity π in hamlets

Exercise 2

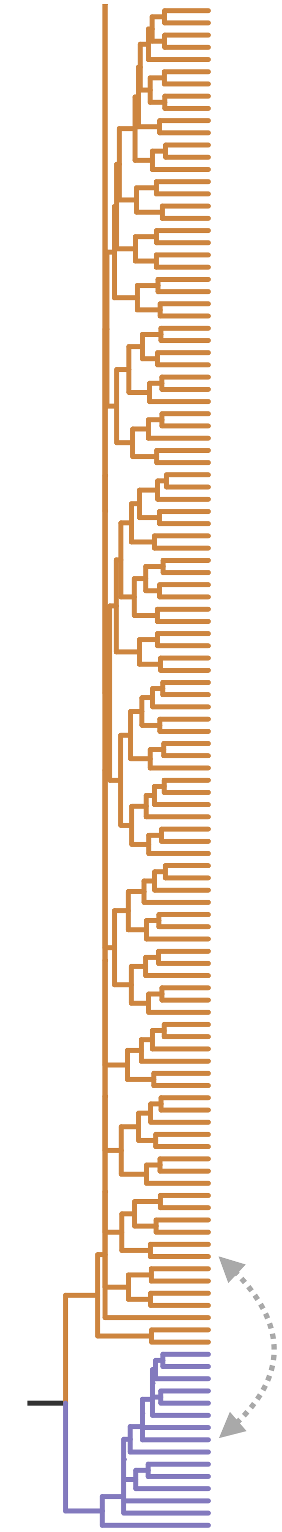


Nucleotide diversity π in hamlets



Elevated π in one population can be explained most easily by having the **longest continuous history** and / or the largest stable N_e

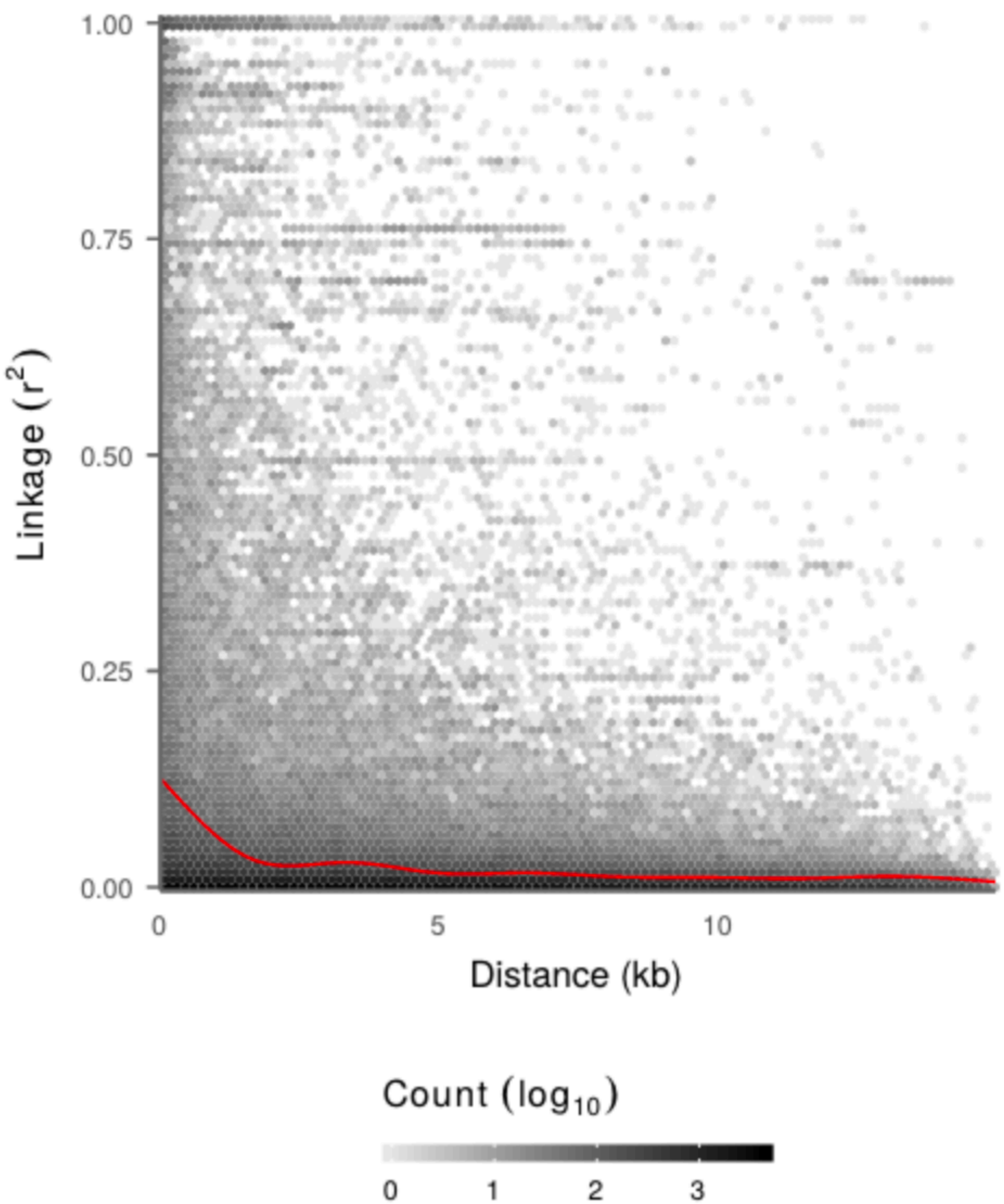
> center of origin



Map data provided by NOAA

Decay of linkage with physical distance

Exercise 2



Hench et al. 2019 (Nat Ecol Evol)

LD statistics

Exercise 2

$D_{AB} = p_{AB} - p_A p_B$

Product of allele frequencies

Haplotype frequency

Coefficient of linkage disequilibrium between two alleles

0 to ±1, but constrained by allele frequencies

$D' = D / D_{max}$

Max value given allele frequencies

D normalized with respect to allele frequencies

0 to ±1, full range (0: no association, ±1: perfect LD)

$$r^2 = \frac{D^2}{p_A\,(1 - p_A)p_B(1 - p_B)}$$

Correlation coefficient of linkage disequilibrium

0 to 1, but constrained by allele frequencies

a.k.a. ρ (*rho*)

Population genomics and genetic diversity

Summary

```
vcftools --gzvcf input.vcf.gz --het --stdout > Het_spain.tsv     # Calculate Fis

for i in pop_*.txt; do                             # Loop over populations to calculate pi
vcftools \
  --gzvcf input.vcf.gz \
  --keep "$i" \
  --window-pi 50000 \
  --stdout > Pi_"$i".tsv
done

vcftools --gzvcf hamlets_maf_lg12.vcf.gz \         # Thin data and calculate R^2
  --thin 5000 \
  --geno-r2 \
  --out ld_thinned
```

Key takeaways

Summary

- **Population genomics** (e.g. genome-wide SNP datasets) provides greater statistical power and finer genomic resolution than traditional markers for detecting evolutionary and demographic processes
- **Elevated F_{IS}** derived from genome-wide SNPs can indicate true inbreeding and increased homozygosity, if confounding factors are carefully considered
- **Nucleotide diversity (π)** measures genetic variation within and among populations, representing the raw material on which selection, genetic drift, and gene flow act
- **Linkage disequilibrium (LD) filtering** helps ensure that analyses are based on approximately independent genetic markers, which is important for unbiased estimates of population structure, relatedness, and demographic parameters

Blackchin guitarfish SNP data

- *Glaucostegus cemiculus*
- 4 individuals from Cabo Verde
- 3 individuals from Spain
- Mapped against Spanish reference genome (2.5 Gb, mean depth 18×, 1503 contigs)
- Genotyped with GATK

guitar_filt_10ctg.vcf.gz

- Chromosome 12 only
- Subset to 36 samples from 6 populations

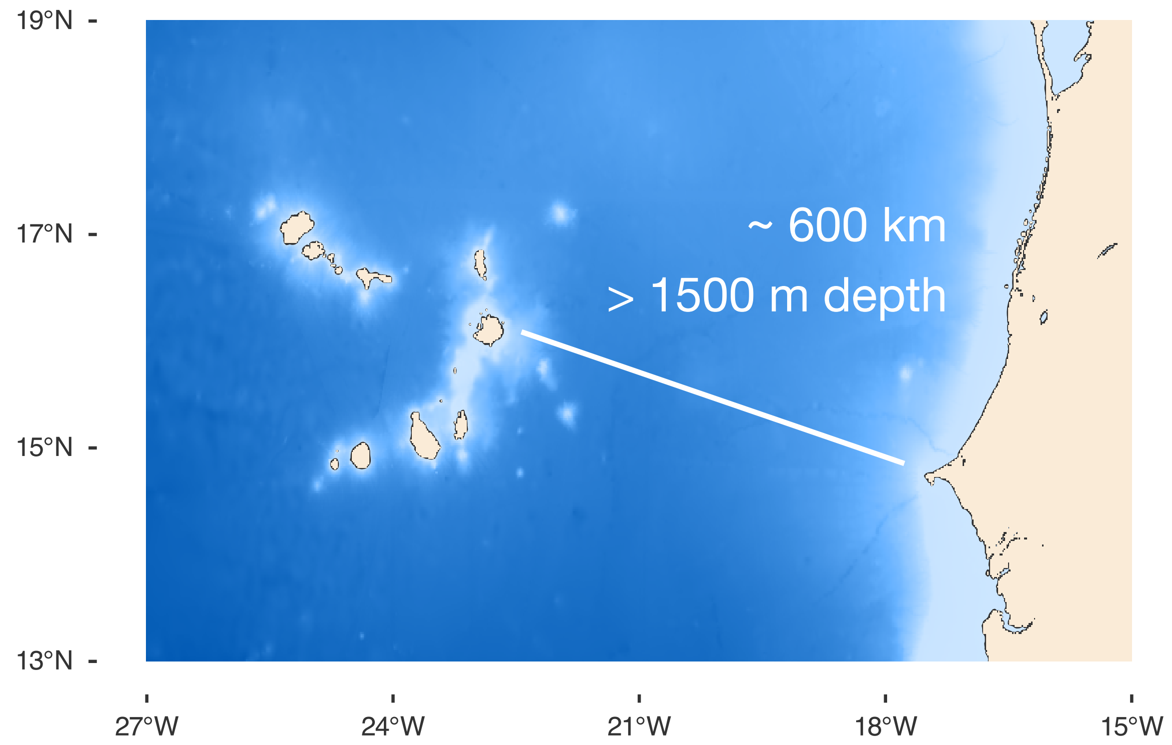


Photo: Jose Martins